

url4 topology — endpoint, node, discovery, addressing, transport — open work

Questions, sketches, spec deltas: the part still moving

STATUS	proposed — owner review; one decision left open in §5; reconciled 2026-09-15 with Kevin's Part A §1.4 refresh, PR #19 commit 2a939bff (draft, unmerged) — see §12 and Appendix D
WORK ITEM	OME-1110
DATE	2026-09-04
OWNER	Sergey Bershadsky (execution and telemetry architecture)
GRAMMAR OWNER	Kevin McDonough (URL4 spec, Parts A/B)
SPEC POINTERS	URL4.ai Specification v0.5 DRAFT — Part A §1.4 at PR #19 (2a939bff), Parts A/B on main, Parts C–I drafts on url4-refactor

Read the core document first; this part assumes its vocabulary.

What this part is

The companion to the core document. It carries what is not yet decided: the questions on authentication (§10), the networking sketch (§11), the alignment with Kevin's spec (§12), the peer-to-peer direction (§13), the spec deltas for Kevin (Appendix A), the follow-up work items (Appendix B) and the vocabulary crosswalk (Appendix D). Read the core first; every term used here is defined there.

10. Deferred, and two questions to move forward

Authentication as a node concern (question, not a decision). Keeping node and endpoint distinct makes a safe place for authentication possible, because it gives the credentials exactly one owner. The spec's inter-node auth (Part H §31, "under active development") propagates per-destination encrypted tokens in `ABC-Auth-Token` headers, and Part B §3.5 names a `URL4-Auth-Token` with a target type. The SDK defers all of it: `url4 serve` ships no authn or authz and asks for a reverse proxy in front. The Engine already has the shape we want: the App mints a per-run capability token and the Runner only carries it.

The idea to test: **the node, through its evaluator, is the only party that holds, validates and issues credentials.**

Endpoints never see a raw credential. An endpoint receives a node-issued session, a capability scoped to one run (`rid`, request tree, purpose, expiry), and uses it for whatever it needs: reading `@` holdings, calling a sibling endpoint, asking the node to fetch an `s3://` source. Outbound, the node forwards the requestor's per-destination tokens and holds the ones addressed to it (Part H §31.5); inbound, the node validates before any endpoint runs. A standalone serverless function is its own node and validates for itself, so the rule holds at every size. The draft already fixes the token's bindings: `rid`, a timestamp with a window of at most 300 seconds, and the destination identity (Part H §31.2).

Questions to answer before this becomes a section:

1. **Session shape.** Is the endpoint-side session the Engine's JWT topic capability generalised (`sub` = run, `iat` window), or the spec's `URL4-Auth-Token`? One token type, or a node-internal one plus the spec's wire one?
2. **Where the identity lives.** `@alice` access control and consent (Part B §5.6.4) need the requestor's identity at the endpoint. Does the session carry it, or does the endpoint ask the node? Part H §33.1 asks the mirror question, whether identity is per server or per endpoint; "per node" is our answer.
3. **Proxy mounts.** For `/claude → url4://beta.example/claude`, which side validates the requestor, and does beta see our identity or a delegated one (Part H §31.2 encrypts to the destination)?
4. **Token destination.** Part H §31 addresses tokens to a node and matches on authority (§31.5). Is the destination always the node, never an endpoint path?
5. **Agent sessions.** The coordinating node issues the `Session-Id` (Part G §28.5). Is that session bound to the run capability, and does the transcript URL it exposes fall under the same access control?
6. **What an endpoint may do with a session.** Only call back into its own node, or reach other nodes directly with a node-minted, destination-bound token?

Recommendation to explore first: node-issued run capability for endpoints (the Engine's pattern), spec §22 tokens between nodes, identity carried in the capability claims. Attribution, consent and audit (Parts E and H) then attach to the same run identity.

A dry-run endpoint on the node (narrowed 2026-09-15: the word is settled, the shape is not). Deferred in the design session; reopened because three later additions gave it inputs it did not have: per-endpoint `accepts` and `emits` (§8), per-endpoint `delivery` and `schemes` (§5, §2), and a node egress that already resolves every target's policy before spending (§11). Today the SDK's `Graph.validate()` checks syntax only and the Engine's preflight checks routes on one node.

Kevin has now defined it. Part A §1.4.3 carries **dry run** — "the process of evaluating an expression without executing the intent, returning only the envelope without a result body" — which is precisely the semantic proposed below. So the what is no longer a question, and this document stops arguing for it. What Part A does not say is how you ask for one, what it costs, or what happens across nodes; questions 1, 2, 4 and 5 below survive unchanged, and question 3 is now a question about the spec's own term rather than about ours.

The shape, in full, so the questions have something to bite on. The draft already obliges an endpoint to do most of it: compute the consumer set by static analysis before resolving anything (Part H §29.2.1), be ready to show a source the full call tree (Part H §30.2), and estimate before executing under a hard budget (Part E §24). Steps 3 and 4 of §11 without step 5: parse, resolve mounts, type-check every edge against `accepts / emits`, check schemes and delivery modes, consult policy and budgets, and return the request tree with per-source verdicts and an estimated cost. Spend nothing. The answer is the envelope (Part D §17) with `sources[]` and `meta` filled in and no `result`; a refused source carries its error code, so "can this node run my expression?" becomes one call with a structured no.

Questions:

1. **Shape.** `Prefer: dry-run` on the ordinary GET (the RFC 7240 pattern we already use for `respond-async`), a `dry_run` protocol parameter beside `q=`, or a separate path? The header keeps the address identical for dry run and run, which suits caching and audit. Part A §1.4.3 defines the operation and says nothing about how it is requested, so this is the open half.
2. **Does a plan spend?** Policy-registry consults and budget reservations are real calls. Is a plan free by definition, with `meta.total_cost` as an estimate from `pricing_version`, or may it reserve?
3. **Is the result an artifact?** A dry-run id returned as a handle, so a later run can say "execute this" and skip re-evaluating. That ties to idempotency (§7) and to the token binding of §11. Note the spec's definition returns "only the envelope without a result body", which does not by itself give the caller anything to hold on to.
4. **Federation.** For a remote subtree, does the node forward the dry run to the other node and merge its answer, or answer only from that node's capabilities document? The first is exact and costs a round trip; the second is instant and approximate. Part A §1.4.3 is silent, and a dry run that silently stops at the first remote edge is worse than one that says it did.
5. **Ownership.** The Engine's preflight admission (OME-880) becomes this endpoint's single-node case. Does the SDK gain `url4 dry-run`, calling the same thing?

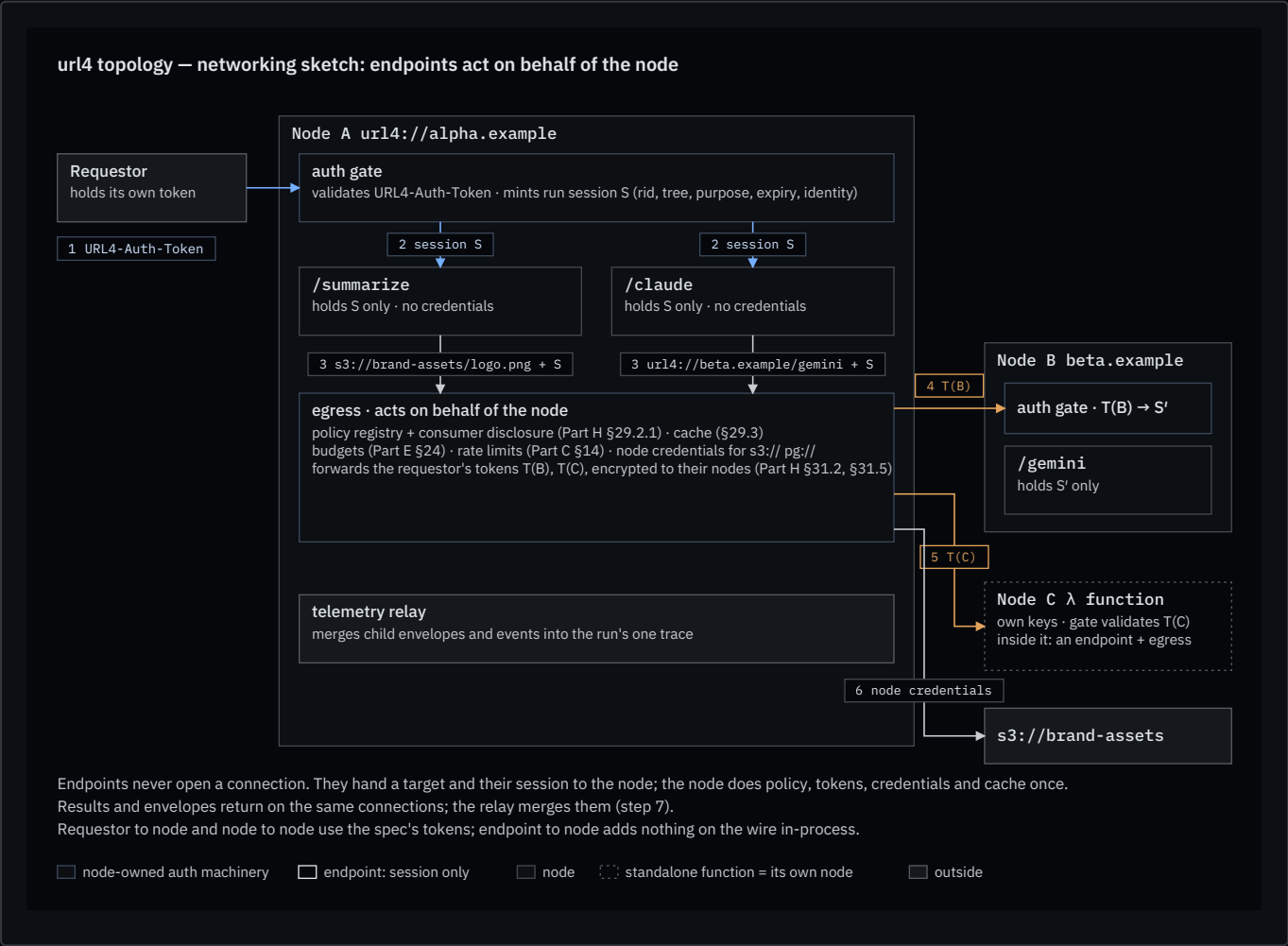
Recommendation to explore first: `Prefer: dry-run` on the same GET, envelope-only answer, cacheable like any GET, forwarded to remote nodes that advertise it and approximated from capabilities where they do not. The word `plan` is retired from this document — the spec's word is **dry run**.

Deferred

- **Swarm** as a protocol noun. Would need membership and trust rules that belong to the unwritten governance parts.
- **Endpoint grades** (processor-only vs evaluator). Rejected: every endpoint evaluates.

11. Networking sketch: endpoints act on behalf of the node

A brainstorm, not a decision. It answers one question from §10 concretely: **how does an endpoint make an outbound request without holding a credential?**



Three candidate mechanisms

	A · NODE EGRESS	B · DELEGATED TOKEN	C · SIDECAR
Who opens the outbound connection	the node	the endpoint, with a node-minted token	a per-endpoint proxy process
Where credentials live	node only; it forwards the requestor's destination-encrypted tokens and holds those addressed to it (Part H §31.5)	the requestor's destination-bound token (Part H §31.2) carried by the endpoint	sidecar only
Policy, disclosure, cache, budgets, rate limits	one place: the node's egress (Part H §29.2.1, §16.3, §20, §31)	at mint time on the node; enforcement split	in the sidecar
Serverless endpoint	nothing to configure	needs network egress and key handling per function	platform-dependent
Cost	an extra hop; large payloads through the node, or by reference (§8)	none	one process per endpoint
What it is today	how the SDK already works: an in-process endpoint fetches through the node's <code>IOLayer</code>	the Engine's per-run JWT, generalised	not built

Recommendation to explore first: A for endpoints, B between nodes. An endpoint never talks to the network. It talks to its node. The node talks to other nodes with the spec's tokens. A standalone serverless function is its own node, so the same two rules cover it: the parent node reaches it node-to-node (B), and inside it the function is an endpoint using its own node's egress (A). C is a deployment shape of A, not a third protocol.

The flow the diagram draws

1. The requestor calls Node A with its `URL4-Auth-Token` (Part B §3.5). Node A's auth gate validates it.
2. The gate mints a **run session S**: `rid`, the request tree, purpose, expiry, the requestor's identity. Every endpoint in the run receives S and nothing else. In-process endpoints get it as an object; command mounts get it in the environment.
3. An endpoint that needs `url4://beta.example/gemini` or `s3://brand-assets/logo.png` hands the target and S to the node's **egress**. It does not open a connection.
4. Egress does the node's work once, in one place: consults the policy registry and discloses consumers (Part H §29.2.1), checks budgets and rate limits (Part E §24, Part C §14), serves from cache when allowed, keyed by requestor identity and consumer set so a shared cache is not a policy bypass (Part H §29.3), and either fetches with the node's own credentials (`s3://`, `pg://`) or forwards the requestor's token **T(B)**, encrypted to Node B, on the sub-request (Part H §31.2, §31.3). The node mints nothing for other nodes: the spec has the originator encrypt every token so that intermediaries cannot read or replace them. What the node mints is internal: the run session S.
5. Node B's gate validates T(B), mints its own session S', and runs `/gemini`. Its result and envelope come back on the same connection.
6. A proxy-mounted standalone function is Node C: same as step 5 with T(C).
7. Node A's telemetry relay merges what came back into the run's one trace (doctrine F1).

What is new on the wire, and what is not

- Requestor to node, and node to node: nothing new. The spec's tokens and `traceparent`.
- Endpoint to node: **nothing on the wire for in-process endpoints**; it is a function call through the node's `IOLayer`, which is what `Url4Node` does today. For out-of-process endpoints (command mounts, containers) an explicit node endpoint is needed, something like `GET /.node/egress?u=<absolute URI>` with `URL4-Session: S`. Its path and shape are open.

Open questions this sketch adds

- Does egress return bytes to the endpoint, or a reference (§8)? For large sources, by reference keeps the node out of the data path.
- The spec binds T(B) to `rid`, a timestamp (≤ 300 s) and the destination (Part H §31.2). Does the node-internal session S need the same three bindings?
- Delegated credentials only target `abc_node` or `http_source` (Part H §31.4). For `s3://` and `pg://` the node's own credentials are the only path; should `target_type` widen, or is node-credential-only the rule?
- Can an endpoint ever be granted direct egress (mechanism B at endpoint level)? Probably only for trusted local mounts, and only by node policy.
- Where does the egress endpoint live for command mounts: a Unix socket, a loopback port, or the node's public origin with S as the credential?

12. Alignment with the spec (2026-09-08 drafts, 2026-09-15 Part A)

The `url4-refactor` branch of `OpenMined/screamingface-design` carries draft Parts C–I (cut 2026-04-28 from the v0.4 text, not yet reviewed) and a v0.4 monolith that rennumbers everything after §20. This document was first written from the v0.2 monolith. Every citation has been re-anchored to the Part numbering. The table records what the drafts changed, and how this document resolved it.

TOPIC	WE WROTE	THE DRAFT SAYS	RESOLUTION
Capabilities document	schema unwritten	Part G §27.2 specifies it: one <code>node</code> , <code>collections</code> , <code>intent_processors</code> , <code>processor_policy</code> , <code>mode_support</code>	§5 rewritten as spec shape + marked additions (<code>path</code> , <code>delivery</code> , <code>schemes</code> , <code>default_processor</code>)
Discovery mechanisms	<code>.well-known</code> vs OPTIONS	<code>.well-known/abc-capabilities</code> SHOULD, <code>Capabilities</code> header MAY (Part G §27.1); no OPTIONS	header added as mechanism C; recommendation moved to A + C
Sub-paths	<code>/name</code> is an endpoint	sub-paths are collections (Part G §27.4); processors are addressed by <code>id</code>	proposed <code>intent_processors[].path</code> ; vocabulary realigned to the spec's Node/Endpoint (2026-09-08)
Mounts vs delegation	<code>proxy</code> forwards a sub-request	<code>processor=</code> delegation, five forms, <code>abc_delegate</code> receives materialised sources (Part G §27.3)	mounts mapped onto processor types (§4); no new mechanism
Async	<code>Prefer: respond-async</code> , <code>Location</code> handle	<code>delivery=async</code> ; 202 body with <code>poll_url</code> (Part C §9.1, §12.5)	adopted; <code>Prefer</code> and <code>Location</code> become aliases
Wrapper switch	<code>Accept: application/json</code> = envelope	<code>Accept</code> negotiates the result's type; envelope is always JSON (Part C §9.1, Part F §25.2)	dedicated <code>application/url4-envelope+json</code> (§6)
Ladder	WS → SSE → sync, ours	two edges only (Part C §11.4); degrade-not-fail is a rule (Part C §10.2)	kept as delta, anchored in §10.2
<code>;accept</code> values	MIME types	short aliases only; registry has no image/audio/video rows (Part F §25.5)	examples fixed; alias rows proposed
Binary results	<code>result.media_type</code>	<code>result.content_type</code> ; data URI in LLM mode, raw for RDS (Part G §26.2.3)	adopted; <code>result.artifact</code> stays a delta
<code>ct_mismatch</code>	fail / convert / pass	four values, default <code>ignore</code> , only with <code>;accept</code> (Part G §26.3)	adopted
Telemetry events, cost event	ours	21 SSE types (Part C §12.5); <code>total_cost</code> scalar, <code>budgets_spent</code> (Part D §17, Part E §24)	kept as additions, roll-up mapped to <code>total_cost</code>
Statelessness	every endpoint keeps nothing	the coordinating node owns an agent session and transcript (Part G §28)	§3: state on the node, endpoints stay stateless
Node-minted tokens	egress mints T(B)	originator encrypts every token; intermediaries forward or hold (Part H §31.2, §31.5)	§11 corrected: node forwards, mints only its internal session
Non-HTTP credentials	node's own	<code>target_type</code> is <code>abc_node</code> <code>http_source</code> only (Part H §31.4)	stated; question for Kevin
Artifacts and proxies	free to forward	flow constraints and redistribution depth (Part H §29.2.2)	cautions added to §4 and §8

TOPIC	WE WROTE	THE DRAFT SAYS	RESOLUTION
Root /	ours	unaddressed; examples use bare <code>abc://node?q=</code> (Part I §41.27); <code>v</code> param outranks the path (Part D §19.4)	kept; tensions listed in §2
Cancel	<code>DELETE</code> on <code>Location</code>	<code>DELETE <poll_url></code> or <code>cancel=<rid></code> (Part C §16.2); <code>cancelled</code> not in the status enum (Part D §17.4)	<code>DELETE <poll_url></code> ; enum gap flagged
Naming	<code>url4://</code> , <code>URL4-*</code>	C-I still <code>abc://</code> , <code>ABC-*</code> ; consolidation is open question 19 (Part I §43)	kept <code>url4</code> ; assumption stated in §1

Confirmed by the drafts without change: the three trust relationships and token bindings (Part H §31), the strict request tree (Part H §29.1), `/name` ensembles on one node (Part I §41.9), a weighted image source (Part I §41.20), consumer disclosure before resolution (Part H §29.2.1), the idempotency wording we dispute (Part C §15.1), and every element of our cancel shape (Part C §16.2).

Kevin's Part A §1.4 refresh (2026-09-15, PR #19 [2a939bfff](#), draft)

A week after the round above, Kevin rewrote Part A §1.4 in response to it. The flat 26-row terminology table became five sub-tables — **1.4.1 Grammar**, **1.4.2 Network**, **1.4.3 Expression Processing**, **1.4.4 Transport**, **1.4.5 General** — and most of Appendix D was absorbed into the spec. Where his text differs from ours, his wins; the table records what we changed in response.

TOPIC	WE WROTE	PART A §1.4 NOW SAYS	RESOLUTION HERE
<code>Mount</code>	ours: <code>local</code> / <code>command</code> / <code>proxy</code> , "not defined" in the spec	§1.4.3, adopted verbatim , alongside the processor types	§1 and §4 cite the spec; delta 1 retired; the stray "N4" annotation becomes delta 15
<code>Evaluator</code> , <code>Evaluation</code>	ours (§1)	§1.4.3, adopted and widened : the evaluator may or may not also be the intent processor	§1 adopts the wider reading
<code>Dry run</code>	our open question, called "plan" (§10)	§1.4.3, defined : evaluate without executing; envelope, no result body	§10 narrowed to shape, cost, federation; "plan" retired as a word
Response ladder	ours, WS → SSE → sync	§1.4.4, adopted , marked "SPEC SECTION NEEDS UPDATING"	§6 rewritten as adopted; delta 4 becomes "close the Part C gap"
<code>websocket</code> delivery	our fourth rung, an addition	§1.4.4, a fourth <code>delivery</code> value beside <code>sync</code> / <code>stream</code> / <code>async</code>	§6; settles Part I question 26 for answers
Envelope switch	ours: <code>application/url4-envelope+json</code>	§1.4.4, adopted , with the <code>meta</code> parameter	§6 marked adopted; delta 7 retired
<code>Scheme adapter</code>	ours (§2)	§1.4.3, adopted , "generalizes Part B §3.5"	§2 cites the spec; says "host's credentials" → delta 13
<code>Degradation</code> , <code>Flow constraints</code> , <code>Attribution</code> , <code>Run handle</code> , <code>Collection</code> , <code>Holdings</code>	ours or implied	§1.4.3–§1.4.5, all now defined	Appendix D becomes a crosswalk rather than a parallel glossary

TOPIC	WE WROTE	PART A §1.4 NOW SAYS	RESOLUTION HERE
Request Tree	strict tree, never a graph (Part H §29.1)	§1.4.3: "the full set of all sources and targets... an evaluator may expand the tree " — strictness dropped	§1 flags the change; delta 14 asks whether it is deliberate. §13's peer-to-peer question depends on the answer
"Host"	deleted from this document 2026-09-08	§1.4.2 adds Host system (a machine); §1.4.4 and the Scheme adapter row then use "host" to mean node	§1 keeps "host" out; delta 13 proposes s/host/node/ in those four rows
Capabilities path	/.well-known/url4-capabilities	§1.4.5 now writes exactly that, where Part G still says abc-capabilities	§5: question 19 resolving in passing
Self-reference, Identity-reference	two rows in the old §1.4	dropped as rows; folded into Holdings	delta 16 asks for confirmation; Holdings also carries Collection 's anchors
Branch	Parts C–I live on url4-refactor	PR #19 targets main , where C–I are stubs, but its rows cite Part C §10.2, Part G §27.3, Part H §31	delta 17: those anchors do not resolve on the branch the PR merges into

Status of that PR. It is a **draft**, one commit, opened 2026-09-15, no reviews. Everything above is pinned to **2a939bff** and may move. Appendix B carries the re-review item.

13. url4 as a network protocol: submit, not call (question)

Raised 2026-09-08. Today a url4 expression is called: one requester sends a GET to one node and waits. The question is whether the same expression can be submitted to a network that resolves it over time, the way a torrent is fetched or a transaction is mined: every requester is a node, capable nodes claim the endpoints an expression names, and the result finds its way back.

What the spec already gives such a network

- **The expression is the address, and it can be content-addressed.** **urn:sha256:** is the accepted form for content addressing (Part I §43, question 1). The canonical rendering of an expression hashes to a stable work id.
- **Submit and collect are already decoupled.** **delivery=async** returns a handle and the result arrives later (Part C §11.3); **resume=<rid>** continues from the last known state (Part C §9.1); results carry **result_version** and **is_final**, so an answer may improve over time (Part C §12.5).
- **Redundant evaluation has a vocabulary.** Quorum and triggers already say how many of N must succeed before an answer counts (Part C §12). Run the same endpoint on several nodes and quorum becomes agreement.
- **Discovery can be a record.** A node's capabilities document (Part G §27.2) is a self-description that can live in a distributed table as well as at **/.well-known**.
- **Every requester is a node.** A local SDK process that evaluates is a node in this document already (§4).
- **Attribution is an incentive model.** Attribution scores and the still-undefined settlement (Part E, Part I §37) are where a Bitcoin-like reward would attach.

What is genuinely new

- **Claiming work.** Which node evaluates a submitted expression, and how two nodes avoid doing the same work, or are made to on purpose.

- **Verifying results without determinism.** A torrent piece is verified by hash; an LLM result cannot be. Candidates: agreement across redundant evaluations (quorum), attested execution (Part I §38), reputation, or accepting unverifiable results for some intents.
- **Consent in an open swarm.** A source may name its permitted consumers and cap redistribution depth (Part H §29.2.2); an anonymous claimer is neither known nor permitted. Either sources opt into open resolution, or the swarm is a closed membership.
- **The context travels.** Submitting an expression means its resolved sources travel to whichever node claims it, the opposite of the "code to the data" posture of RDS mode (Part A §1.1).
- **Liveness and economics.** Who guarantees an expression is ever resolved, and who pays. Part I §37 is TBD.

A sketch to argue with. The requester canonicalises the expression and publishes `urn:sha256:<hash>` with the expression to the network. Nodes whose capabilities cover an endpoint the expression names claim that endpoint and evaluate it against their own holdings and credentials, as §11 describes for one node. Each returns a signed envelope keyed by the hash. The requester's `poll_url` becomes a lookup of that key; quorum decides when the answer is good enough; `is_final` closes it. Nothing in the grammar changes; the transport changes from point-to-point to publish-and-claim.

Questions

1. Is this a fourth delivery mode (`delivery=network`) or a different transport beneath the same three?
2. Which verification candidate is acceptable for which kind of intent?
3. Can consent and flow constraints be expressed for an open set of claimers, or is membership required?
4. Does the requester's evaluator keep fan-out and reduce, with the network resolving only leaf endpoints? That keeps §11's egress intact and makes the swarm a resolver, not an orchestrator.

Appendix A — proposed spec deltas for Kevin

Each item names the anchor and the change. All are proposals. Re-triaged 2026-09-15 against Part A §1.4 at PR #19 (2a939bff, draft).

Landed in PR #19

Kept as the record of what was proposed and taken. Nothing is asked here.

#	WAS	WHERE IT LANDED
1	Part A §1.4 — endpoint kinds. Name the binding kinds <code>local</code> <code>command</code> <code>proxy</code> .	§1.4.3 <code>Mount</code> , verbatim . Superseded in one respect: no rename was needed — Node and Endpoint stayed, and §1.4.2 added <code>Node address</code> and <code>Endpoint path</code> instead.
4	Part C §10.2, §11.4 — ladder and floor. WebSocket → SSE → sync in one round trip; sync the only MUST; answering below the ask is a reported degradation.	§1.4.4 <code>Response ladder</code> and <code>Degradation</code> , plus <code>websocket</code> as a fourth <code>delivery</code> value. Still open below — Part C itself is unchanged, as Kevin's own row notes.
7	Part D §17 — envelope for sync callers. <code>application/url4-envelope+json</code> selects the envelope; <code>meta</code> sets the depth.	§1.4.4 <code>Envelope</code> . The <code>telemetry</code> block and the <code>meta=summary</code> default are not stated there; delta 12 still stands.
9	Part B §3.5 — other schemes. Any non-HTTP scheme is a read through a node-mounted adapter, advertised in capabilities.	§1.4.3 <code>Scheme adapter</code> , "generalizes Part B §3.5". Says "host's own credentials"; see delta 13.

#	WAS	WHERE IT LANDED
10	Part G §27.2 — capabilities document. Add <code>delivery</code> , <code>schemes</code> ; <code>url4_*</code> naming.	§1.4.5 <code>Capabilities</code> document now lists delivery modes and schemes and writes <code>/well-known/url4-capabilities</code> . <code>intent_processors[].path</code> , <code>inline_max_bytes</code> and <code>default_processor</code> are still open below.

Still open

- **2. Part B §3.1.1, Part D §19 — root.** `url4://node` $\equiv$ `url4://node/`; `/` is the node's default endpoint at the current version; `/v1` is a version alias. Your own examples already do this (Part I §41.27). Also resolve §19.1 (path primary) against §19.4 (`v` outranks path). Untouched by PR #19.
- **3. Part C §11.2 — bindings.** `delivery=stream` is SSE on the same GET, requested with `Accept: text/event-stream` (§11.2 names SSE but no media type). Part A §1.4.4 now says `stream` is "SSE events on the same GET", so this is a Part C catch-up.
- **4. Part C §10.2, §11.4 — make Part C match Part A.** §1.4.4 now carries the four delivery modes and the full ladder; Part C §11 still describes three modes and Part C §11.4 two degradation edges. Your own §1.4.4 row is annotated SPEC SECTION NEEDS UPDATING. Add a `delivery` list per processor to the capabilities document at the same time.
- **5. Part C §15.1 — idempotency.** Replace "the protocol does not guarantee idempotency" with: GET is idempotent per RFC 9110 §9.2.2; results are not guaranteed deterministic. §15.2's `budget_exceeded` row stays as the stated exception.
- **6. Part C §16 — cancellation.** Take your first option: `DELETE <poll_url>`; drop `cancel=<rid>`. Terminal state `cancelled`; SSE event `request.cancelled`; propagates to in-flight children; partial result when quorum was met. Add `cancelled` to the status enum in Part D §17.4 and to the state machine in Part C §13.4.
- **8. Part F §25, Part G §26 — typed payloads.** Add image, audio and video rows with aliases (`png`, `jpeg`, `wav`, `mp4`) to the §25.5 registry; extend §26.2.3's data-URI rule to results (`result.content + result.content_type`); add `result.artifact` for by-reference results above an endpoint-declared `inline_max_bytes`, subject to §29.2.2 flow constraints. Part E: say how weights and budgets apply to non-text sources. Slot §41.24 is free for a multimodal example.
- **11. Part H §31.4 — credential targets.** Either widen `target_type` beyond `abc_node | http_source`, or state that non-HTTP schemes are resolved with the node's own credentials only (§2). §1.4.3's `Scheme adapter` now assumes the latter without saying so.
- **12. Part D §17.2.2, §20 — additions must respect the Hybrid Rule and propagated.** Our `telemetry` and `session` blocks are present-and-null at `summary`; unknown fields pass through unaggregated.

New, raised by PR #19

- **13. Part A §1.4.2, §1.4.4 — host vs node.** `Host system` is welcome in §1.4.2 as a deployment word: a machine supporting one or more nodes. But §1.4.4 `Delivery mode` ("the negotiated method by which a **host** responds") and `Response ladder` ("the mechanism by which a **host** responds with the richest delivery mode"), and the §1.4.3 `Scheme adapter` row ("using the **host's** own credentials"), all use "host" where the actor is the **node**. It is the node that negotiates delivery, picks a rung and holds scheme credentials; a host system negotiates nothing, and on a machine running two nodes the sentence has no referent. Proposal: `s/host/node/` in those rows and keep `Host system` to §1.4.2 alone.
- **14. Part A §1.4.3 — Request Tree lost its strictness.** The new row reads "the full set of all **sources** and **targets** defined in an **expression**. An **evaluator** may expand the tree during **source resolution** and sub-expression **evaluation**." Part H §29.1 says the request tree is strictly a tree, never a graph, and this document relies on that in §3, §7 and §13. Is the relaxation deliberate — is a node allowed to converge two branches onto one sub-expression — or is "expand" only meant to say the tree is discovered lazily? The answer decides whether §13's publish-and-claim model is a transport change or a model change.

- **15. Part A §1.4.3 — a doctrine reference leaked into the spec.** The `Mount` row annotates `command` as "(subprocess, N4)". N4 is an item in our internal doctrine skill (`.claude/skills/url4-engine/SKILL.md`), not a spec anchor; a reader of Part A cannot resolve it. Drop it, or replace it with Part H §36, which is the rule that actually governs command mounts.
- **16. Part A §1.4.1 — Holdings anchors, and two dropped rows.** `Holdings` carries Part B §5.3, Part G §27.4, which are `Collection` 's anchors; holdings and the `@` token are Part B §5.6. Separately, `Self-reference` and `Identity-reference` were rows in the old §1.4 and are now gone, apparently folded into `Holdings` — please confirm that is intended, because `@alice` carries an access-control and consent rule (Part B §5.6.4) that the single `Holdings` line does not state.
- **17. Branch and anchors.** PR #19 targets `main`, where Parts C–I are "Not yet written" stubs. Its new rows nonetheless cite Part C §10.2, Part C §11, Part C §12.5, Part G §27.3 and Part H §31 — anchors that resolve only on `url4-refactor`. Either the drafts merge first, or Part A §1.4 ships with references a reader of `main` cannot follow.
- **18. Part A §1.4 — typos and markup.** `Inter-host tone` should be `Inter-host token` (and, per delta 13, inter-node token). `Request identiifer` $\times 2 \rightarrow$ `identifier`. `Extention` $\rightarrow$ `extension`. Unclosed bold in "nested ****Expression***" (§1.4.1 `Source`) and "one or more ****nodes***" (§1.4.2 `Host system`). "one **ore** more" (§1.4.3 `Resolution`). "without executing **then** intent" (§1.4.3 `Dry run`). "advertising **a its** collections" (§1.4.5 `Capabilities document`). A stray `.` before "Generalizes" in `Scheme adapter`.

Appendix B — follow-up work items

To file after owner review, one per landing.

LANDING	ITEM
screamingface-engine	Node surface phase 1: discovery per §5, <code>GET /?q=</code> with SSE, <code>DELETE</code> on the run handle
screamingface-engine	Model endpoints reachable over HTTP (phase 2); standalone model functions, proxy-mounted (phase 3)
screamingface-engine	Rename the <code>URL4-Capability</code> JWT header to avoid the "capabilities" collision
url4-sdk	Delivery negotiation in <code>Url4Node.asgi()</code> : <code>Upgrade / Accept</code> handling, <code>application/url4-envelope+json</code> , SSE body, <code>delivery=async</code> with <code>poll_url</code> , optional WebSocket answer; evaluator-side single-request ladder in <code>HttpIOLayer</code>
url4-sdk	Capabilities document and/or OPTIONS responder, after the §5 decision; advertise <code>delivery</code> per endpoint
url4-sdk	<code>url4 serve</code> default path <code>/</code> ; <code>Client</code> default path <code>/</code> ; <code>/v1</code> alias
url4-sdk	<code>proxy</code> mount kind in <code>url4.toml</code>
url4-sdk	Scheme adapters (<code>s3://</code> , <code>pg://</code> , <code>sqlite://</code> , ...) as <code>IOLayer</code> adapters registered by scheme; <code>[schemes]</code> in <code>url4.toml</code> ; <code>schemes</code> in capabilities
url4-sdk	Typed payloads: bytes + media type through <code>IOLayer / FetchRequest</code> , <code>;accept / ct_mismatch</code> enforcement, <code>result.artifact</code> by-reference fetch
screamingface-engine	Non-text processors over aigateway (image, speech); artifact store as the by-reference path; <code>accepts / emits</code> in capabilities
url4-sdk	<code>Url4Node</code> $\rightarrow$ node naming retrofit with a deprecation alias

LANDING	ITEM
repo	Doctrine skill synced in this unit (T1, N1, F2, F4, term table)
repo	Re-review this document when PR #19 leaves draft: every "adopted" claim here is pinned to 2a939bff and the PR has had no review yet
Kevin	Review Appendix A — deltas 2, 3, 4, 5, 6, 8, 11, 12 still open; 13–18 raised by PR #19 itself

Appendix D — Vocabulary crosswalk

Until 2026-09-15 this appendix was a glossary of its own, written because Part A §1.4 was a flat 26-row table that left most of this document's vocabulary undefined. PR #19 changed that: §1.4 is now five sub-tables and **most of these terms are the spec's own**. A second glossary would only drift from it, so what remains is the crosswalk — for each term, where it lives in Part A §1.4 as of [2a939bff](#), and whether anything is still owed.

Read the definitions in the spec. Read this to know which of them came from here, which changed on the way, and which are still only ours.

Status key. adopted — in Part A §1.4, saying what we meant · changed — adopted, but the meaning moved; see the delta · spec's own — the spec's word before this document existed · ours — not in the spec; this document is the only definition.

Adopted from this document

TERM	PART A §1.4	STATUS	STILL OWED
Mount	§1.4.3	adopted	The row annotates <code>command</code> with "N4", our doctrine numbering (delta 15).
Evaluator, Evaluation	§1.4.3	adopted, widened	The evaluator "may or may not" also be the intent processor — wider than §1's reading, and we take the wider one.
Dry run	§1.4.3	adopted	The operation is defined; how you ask for one, what it costs and how it federates are open (§10).
Response ladder	§1.4.4	adopted	Part C §11.4 still lists two edges. Kevin's own row says SPEC SECTION NEEDS UPDATING (delta 4).
Degradation	§1.4.4	adopted	—
Scheme adapter	§1.4.3	adopted	Says "the host's own credentials"; it is the node's (delta 13).
Envelope	§1.4.4	adopted	Carries the <code>Accept: application/url4-envelope+json</code> switch and <code>meta</code> . The <code>telemetry</code> block and the <code>meta=summary</code> default are still deltas (12).
Run handle, <code>poll_url</code>	§1.4.4	adopted	—
Flow constraints	§1.4.5	adopted	—
Attribution	§1.4.3	adopted	—
Delivery mode	§1.4.4	adopted	<code>websocket</code> is now the fourth value; <code>sync</code> the only MUST.
Request tree	§1.4.3	changed	Strictness is gone: "an evaluator may expand the tree". Part H §29.1 says strictly a tree. §3, §7 and §13 depend on which is true (delta 14).

TERM	PART A §1.4	STATUS	STILL OWNED
Inter-node token	§1.4.4	changed	Appears as "Inter-host tone " — a typo, and "host" for "node" (deltas 13, 18).

The spec's own, before this document

TERM	PART A §1.4	NOTE
Expression, Source, Intent	§1.4.1	Intent is newly spelled out: a prompt, a code pointer, a URI, or an expression that must resolve to one of the first two.
Collection, Holdings	§1.4.1	Holdings is new as a row and folds in the old Self-reference / Identity-reference ; its anchors are Collection 's (delta 16).
Broadcast, Expansion	§1.4.1	Unchanged.
Node, Endpoint	§1.4.2	Endpoint is now an interface, named by an endpoint path — see below.
Requestor, Target	§1.4.3	Target is now explicitly node address + endpoint path.
Intent processor	§1.4.3	Unchanged in substance.
Resolution, Execution	§1.4.3	Unchanged.
Terminal state, Quorum, Trigger	§1.4.3	Trigger now says intermediate result.
Agent session	§1.4.3	The basis for the statelessness carve-out in §3.
Policy registry, Data owner, Purpose, Jurisdiction, Settlement, Attestation	§1.4.5	Governance vocabulary; unchanged.
URL4-aware, Non-URL4 source	§1.4.5	Unchanged.
Capabilities document	§1.4.5	Now written /.well-known/url4-capabilities and naming the Capabilities header (§5).
Idempotent	Part C §15	Not in §1.4. Wording still disputed (delta 5).

New in PR #19, adopted here

TERM	PART A §1.4	HOW THIS DOCUMENT USES IT
Node address	§1.4.2	§2: scheme + RFC 3986 authority.
Endpoint path	§1.4.2	§1, §2, §3: /claude is the path; the endpoint is the interface it names.
Host system	§1.4.2	Not used. A machine supporting one or more nodes is a fine deployment word and has no protocol role here; §1 and delta 13 explain why "host" stays out.

Still only ours

Nothing in Part A §1.4 defines these. Each is a live proposal with a delta behind it.

TERM	DEFINITION	WHERE
Typed payload	Every edge carries a value named by its media type; text is the default. Sources declare by <code>Content-Type</code> , results by <code>result.content_type</code> .	§8, delta 8
Artifact	A large result returned as an <code>https://</code> URL the next endpoint fetches as data, above an endpoint-declared <code>inline_max_bytes</code> ; inherits flow constraints.	§8, delta 8
Run session	A node-issued capability scoped to one run (<code>rid</code> , tree, purpose, expiry, identity) that endpoints carry instead of credentials.	§10, §11
Egress	The node component that performs every outbound request for its endpoints: policy, disclosure, cache, budgets, credentials, token forwarding.	§11
Telemetry signals	Logs, spans (tokens live here), and <code>cost.usage</code> events (money lives here). In-band in every mode; OTLP is the durable copy.	§7, delta 7
OPTIONS discovery	One answer per endpoint, mechanism B of §5. Appears nowhere in the spec, and PR #19 did not add it.	§5